On the Summability Problem of Multivariate Rational Functions in the Mixed Case

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Summability Problem

Problem. Determine whether $f(x_1,\dots,x_n)$ can be written as

$$f = \Delta_{x_1}(g_1)+\cdots+\Delta_{x_n}(g_n),$$

Application.

$$\sum_{i=0}^{n}\sum_{j=0}^{n}\binom{i+j}{i}^2\binom{4n-2i-2j}{2n-2i} = (2n+1)\binom{2n}{n}^2.$$

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where $f, g_1, \ldots, g_n \in \mathbb{F}(x_1,\ldots,x_n)$ with $\mathbb{F}$ a field of characteristic 0.

q-Summability Problem

Let $q \in \mathbb{F}$ such that $q^i \neq 1,\ \forall i \in \mathbb{Z}$.

Problem. Determine whether $f(x_1, \ldots, x_n)$ can be written as

$$f = \Delta_{q,x_1}(g_1) + \cdots + \Delta_{q,x_n}(g_n).$$

where Δ_{q,x_i} is the q-difference operator w.r.t. x_i.

Application. Evaluate

$$\sum_{(z_1,\ldots,z_n)\in\Omega} f(x_1, \ldots, x_n), \quad \text{where } x_i = q^{z_i} \text{ and } \Omega \subseteq \mathbb{Z}^n.$$

Mixed Summability Problem

Motivation. Let $n = k + m$. Evaluate a multiple sum:

$$\sum_{(y_1,\ldots,y_k,z_1,\ldots,z_m)\in\Omega} f(\underbrace{y_1,\ldots,y_k,q^{z_1},\ldots,q^{z_m}}_{x_1,\ldots,x_n}), \quad \text{where } \Omega \subseteq \mathbb{Z}^n.$$

Shift operators.

$$\theta_{x_i}(f(x_1,\ldots,x_n)) := \begin{cases} f(x_1,\ldots,x_i+1,\ldots,x_n), & \text{if } x_i \in \{y_1,\ldots,y_k\}; \\ f(x_1,\ldots,q\cdot x_i,\ldots,x_n), & \text{if } x_i \in \{q^{z_1},\ldots,q^{z_m}\}. \end{cases}$$

Problem. Determine whether f can be written as

$$f(x_1,\ldots,x_n) = \sum_{i=1}^{n} \Delta_{\theta_{x_i}}(g_i(x_1,\ldots,x_n)).$$

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Problem. Determine whether

$$f(x_1,\ldots,x_n) \text{ is } (\theta_{x_1},\ldots,\theta_{x_n})\text{-summable in } \mathbb{F}(x_1,\ldots,x_n).$$

Univariate Rational Summation

Problem. Given a sequence $f(k) \in \mathbb{F}(k)$, find $g(k) \in \mathbb{F}(k)$ such that

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Note that if $f \in \mathbb{F}[k]$, then one can always find such g.

Solution. Abramov (1975), Paule (1995), ...

Examples.

$$\sum_{0 \le k < n} k = \frac{n(n-1)}{2} \qquad \sum_{1 \le k \le n} \frac{1}{k(k+1)} = \frac{n}{n+1}$$

$$\sum_{0 \le k < n} k^2 = \frac{n(n-1)(2n-1)}{6}$$

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Abramov's Algorithm

Additive decomposition. Given $f \in \mathbb{F}(k)$, compute $g \in \mathbb{F}(k)$ and polynomials $a, d \in \mathbb{F}[k]$ such that

$$f = \Delta_k(g) + \frac{a}{d}, \text{ where } \deg_k(a) < \deg_k(d)$$

and d has the minimal degree among all possible choices, satisfying

$$\frac{d(k+i)}{d} \notin \mathbb{F}, \quad \text{for all } i \in \mathbb{Z}.$$

Summability criterion.

$$f \text{ is } \sigma_k\text{-summable in } \mathbb{F}(k) \quad \Leftrightarrow \quad a = 0.$$

Bivariate Rational Summation

Problem. Given $f \in \mathbb{F}(x,y)$, determine whether f can be written as

$$f = \Delta_x(g) + \Delta_y(h), \text{ where } g,h \in \mathbb{F}(x,y).$$

Additive decomposition w.r.t. y.

$$f = \Delta_x(g) + \Delta_y(h) + \sum_{j=0}^{J} \sum_{i=0}^{I} \frac{a_{i,j}}{d_i^j}, \text{ where } \deg_y(a_{i,j}) < \deg_y(d_i)$$

with d_i's being irreducible and not shift equivalent with each other.

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Theorem (ChenSinger2014, HouWang2015).

$$f \text{ is } (\sigma_x, \sigma_y)\text{-summable in } \mathbb{F}(x,y)$$

$$\Updownarrow$$

$$\text{each } \frac{a_{i,j}}{d_i^j} \text{ is } (\sigma_x, \sigma_y)\text{-summable in } \mathbb{F}(x,y)$$

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$$d_i = P_i(\alpha_i x + \beta_i y),\ \alpha_i, \beta_i \in \mathbb{Z},\ P_i \in \mathbb{F}[z] \text{ and } \sigma_x^{\beta_i}\sigma_y^{-\alpha_i}(a_{i,j}) = a_{i,j}.$$

Multivariate Rational Summation

Additive decomposition w.r.t. x_1. Let $\mathbf{x} := x_1, \ldots, x_n$ and $f \in \mathbb{F}(\mathbf{x})$,

$$f = \Delta_{x_1}(g_1) + \cdots + \Delta_{x_n}(g_n) + \sum_{i=1}^{I} \sum_{j=1}^{J_i} \frac{a_{i,j}}{d_i^j}, \text{ where } \deg_{x_1}(a_{i,j}) < \deg_{x_1}(d_i)$$

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Theorem. Both G_p and G/G_p are free and $\operatorname{rank}(G_p) \leq n-1$.

Summability Criteria

Theorem (ChenDuFang2025). Let $f = a/d^s$ with irreducible d and $\deg_{x_1}(a) < \deg_{x_1}(d)$. Let $\tau_1, \ldots, \tau_r$ be a basis of G_d with $r < n$.

$$f \text{ is } (\sigma_{x_1}, \ldots, \sigma_{x_n})\text{-summable in } \mathbb{F}(\mathbf{x})$$

$$\Updownarrow$$

$$a = \tau_1(b_1) - b_1 + \cdots + \tau_r(b_r) - b_r \text{ for some } b_i \in \mathbb{F}(\mathbf{x})$$

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$\Updownarrow$ via a difference transformation ϕ

$\phi(a)$ is $(\sigma_{x_1}, \ldots, \sigma_{x_r})$-summable in $\mathbb{F}(\mathbf{x})$

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Strategy. Reduce the problem to the case of fewer variables.

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Example. Let $f := 1/d$ with $d := x_1^s + \cdots + x_n^s$ where $n > 1, s \geq 1$.

We have

$$f \text{ is } (\sigma_{x_1}, \ldots, \sigma_{x_n})\text{-summable in } \mathbb{F}(\mathbf{x}) \quad \Leftrightarrow \quad s = 1.$$

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Example. Let $f := 1/d$ with $d := x_1^s + \cdots + x_n^s$ where $n > 1, s \geq 1$.

If $s > 1$, then $G_d = \{\mathbf{1}\}$.

$$\text{So } f \text{ is not } (\sigma_{x_1}, \ldots, \sigma_{x_n})\text{-summable in } \mathbb{F}(\mathbf{x}).$$

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If $s = 1$, then

$$G_d = \langle \tau_1, \ldots, \tau_{n-1} \rangle \quad \text{and } 1 = \tau_1(x_1) - x_1,$$

where $\tau_i = \sigma_{x_i} \sigma_{x_{i+1}}^{-1}$. So f is summable in $\mathbb{F}(\mathbf{x})$.

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Example. Let $f := 1/d$ with $d := x_1^s + \cdots + x_n^s$ where $n > 1$, $s \geq 1$.

If $s = 1$, then

$$\frac{1}{x_1 + \cdots + x_n} = \Delta_{x_1}\left(\frac{x_1}{x_1 + \cdots + x_n}\right) + \Delta_{x_2}\left(\frac{-x_1 - 1}{x_1 + \cdots + x_n}\right).$$

Multivariate Rational Summation in the Mixed Case

Goal.

- Reduction to the case of fewer variables.
- Unified framework for mixed shift case.

Solution (What is new?).

- Generalize the problem: Given $f \in \mathbb{F}(\mathbf{x})$ and $\mathbf{c} \in \mathbb{F}^n$, determine whether f is $(c_1\theta_{x_1}, \ldots, c_n\theta_{x_n})$-summable in $\mathbb{F}(\mathbf{x})$.
- Distinguish the normal and special cases of the denominator d.
- Direct decomposition of the isotropy group $G_d = G_d^{\sigma} \oplus G_d^{\tau}$.
- Construct a combined difference transformation ϕ.

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Normal polynomials. Let p denote a polynomial in x_1 over $\mathbb{F}(\check{\mathbf{x}})$. Define that p is normal w.r.t. θ_{x_1} if $\gcd(p, \theta_{x_1}^{\ell}(p)) = 1,\ \forall \ell \in \mathbb{Z} \setminus \{0\}$.

Special polynomials. p is called special w.r.t. θ_{x_1} if p is not normal. For example, we have x_1 is special w.r.t. τ_{q,x_1} since $\tau_{q,x_1}(x_1) = q \cdot x_1$.

Isotropy group. Let $G = \langle \theta_{x_1}, \ldots, \theta_{x_n} \rangle$. For $p \in \mathbb{F}[\mathbf{x}] \setminus \mathbb{F}$, define

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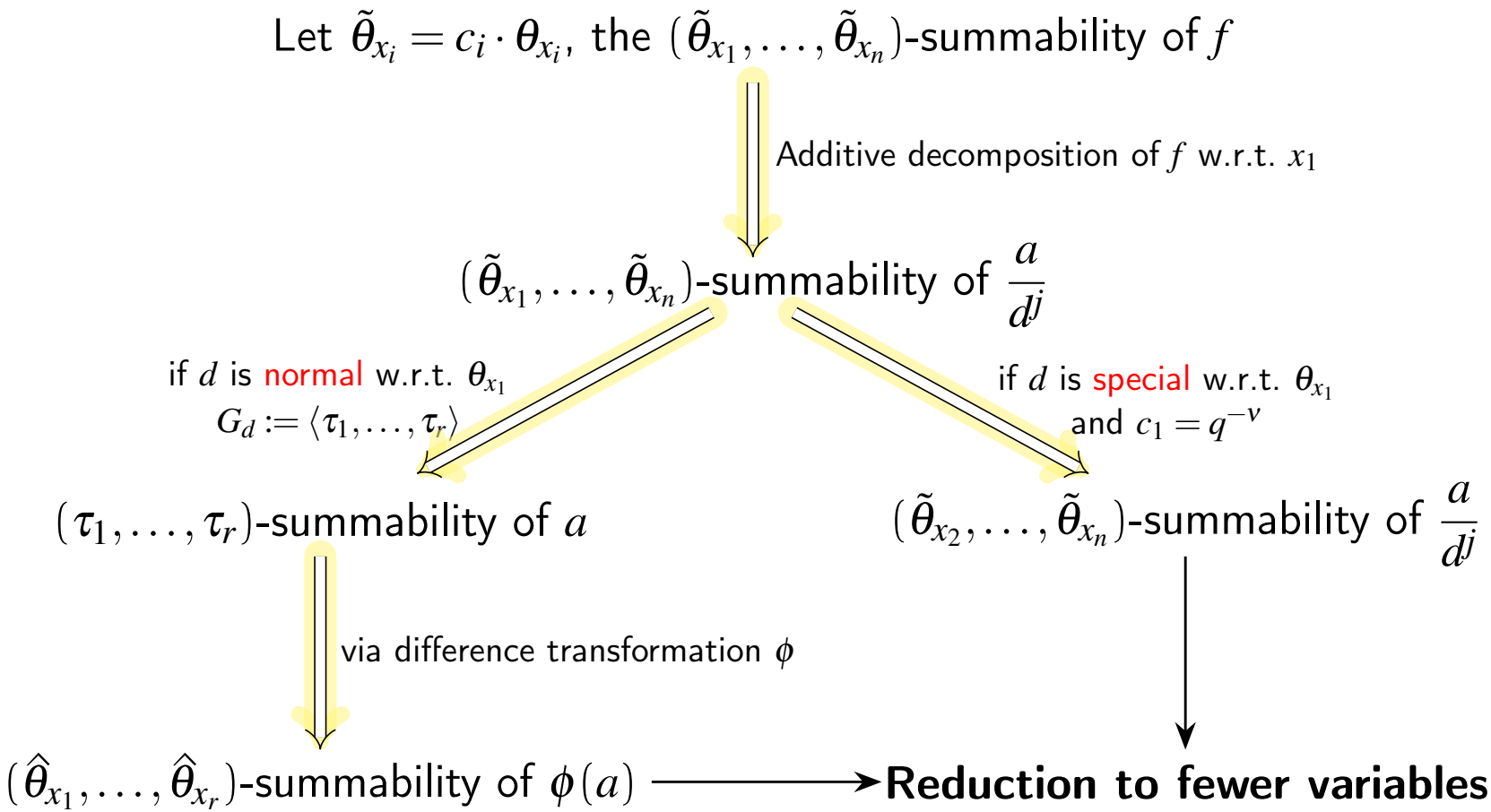
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A Recursive Strategy



Example

Determine whether f is $(\tau_{q,x_1},\ldots,\tau_{q,x_n})$-summable in $\mathbb{F}(\mathbf{x})$, where

$$f := \frac{1}{x_1^s+\cdots+x_n^s} \text{ whith } s,n \in \mathbb{N}^+.$$

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When $n>1$, $G_d =< \tau := \tau_{q,x_1}\cdots\tau_{q,x_n} >$ with $\tau(d) = q^s\cdot d$. Since

$$1 = q^{-s}\tau\left(\frac{1}{q^{-s}-1}\right) - \frac{1}{q^{-s}-1},$$

we have $f = \sum_{i=1}^n \Delta_{\tau_{q,x_i}}(g_i)$, where

$$g_i := \tau_{q,x_{i+1}}\circ\cdots\circ\tau_{q,x_n}\left(\frac{1}{(q^{-s}-1)(x_1^s+\cdots+x_n^s)}\right).$$

Applications of Evaluating Sums

We show that $S(q)$ converges for $0 < |q| < 1$, where

$$S(q) := \sum_{n,k,m \geq 1} \left(\frac{q^{k+m}}{[n+k+m]_q} - (1-q)q^{k+m} \right) \text{ with } [N]_q := \frac{1-q^N}{1-q}.$$

Let $(x,y,z) := (q^n, q^k, q^m)$. Then considering the first sum we have

$$\frac{(1-q)yz}{1-xyz} = \Delta_{\tau_{q,x}} \left(\frac{yz}{1-q^{-1}xyz} \right) - \Delta_{\tau_{q,y}} \left(\frac{yz}{1-q^{-1}xyz} \right).$$

In conclusion, we have

$$S(q) = 2\left(\frac{q}{1-q}\right)^2 - \sum_{k,m \geq 1} \frac{q^{k+m}}{1-q^{k+m}} - q \sum_{m \geq 1} \frac{q^m}{1-q^m}.$$

Conclusion and Future Work

Conclusion. Determine whether $f \in \mathbb{F}(x_1, \ldots, x_n)$ can be written as

$$f = \Delta_{\theta_{x_1}}(g_1) + \cdots + \Delta_{\theta_{x_n}}(g_n),$$

where $g_i \in \mathbb{F}(x_1, \ldots, x_n)$. If this is the case, compute the g_i's.

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Future work.

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Thank you!